

Lecture 09 - Managing environments with conda

Introduction to Data Science and Programming

September 23th, 2025

Installing packages

(That haven't been installed with the Anaconda distribution)

Let's say we want to use the package “beautifulsoup4”

Command line to see which packages are available:

```
conda list
```

Installing packages

(That haven't been installed with the Anaconda distribution)

Command line:

`conda install beautifulsoup4` or `pip install beautifulsoup4`

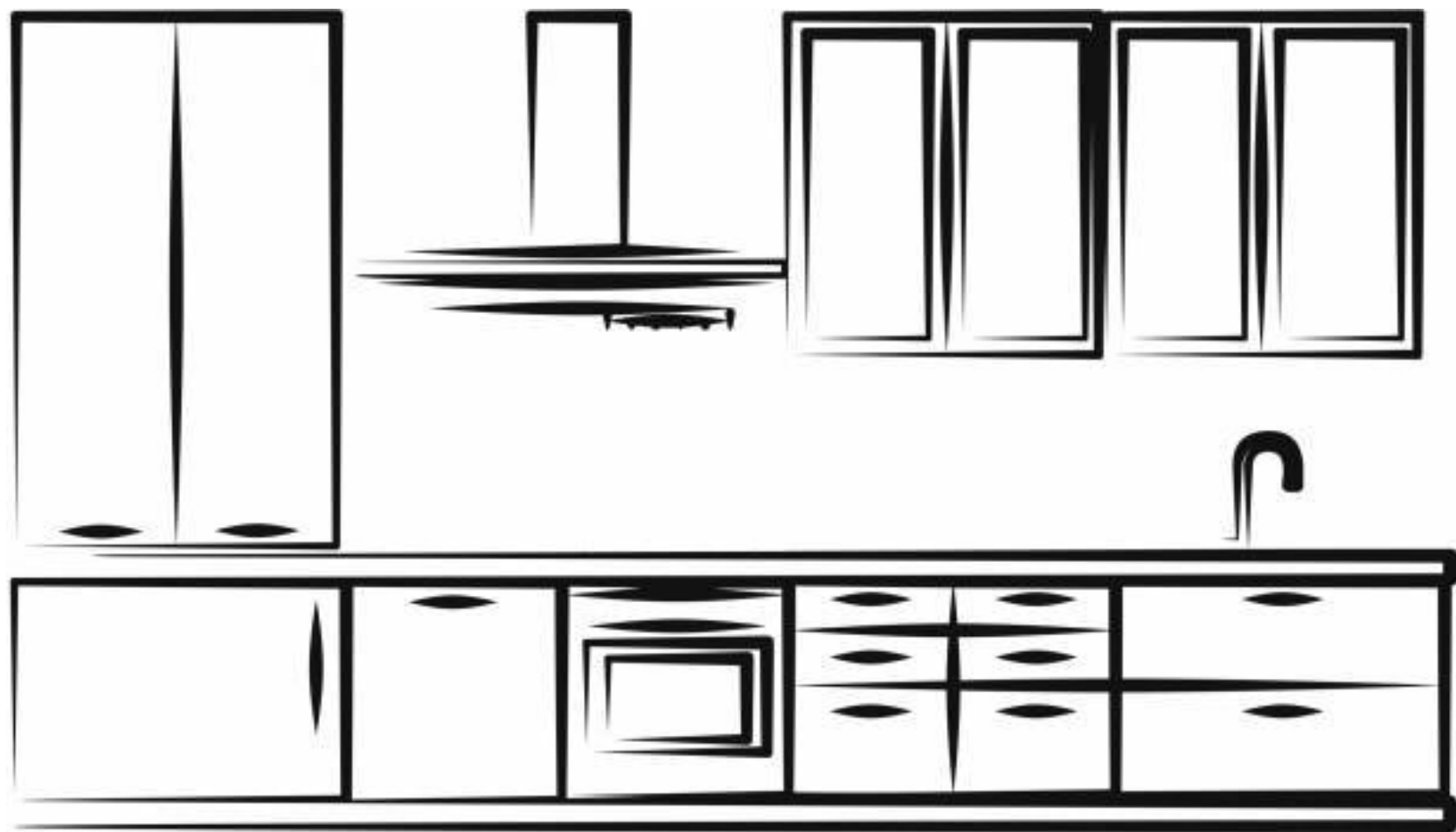
Installing packages

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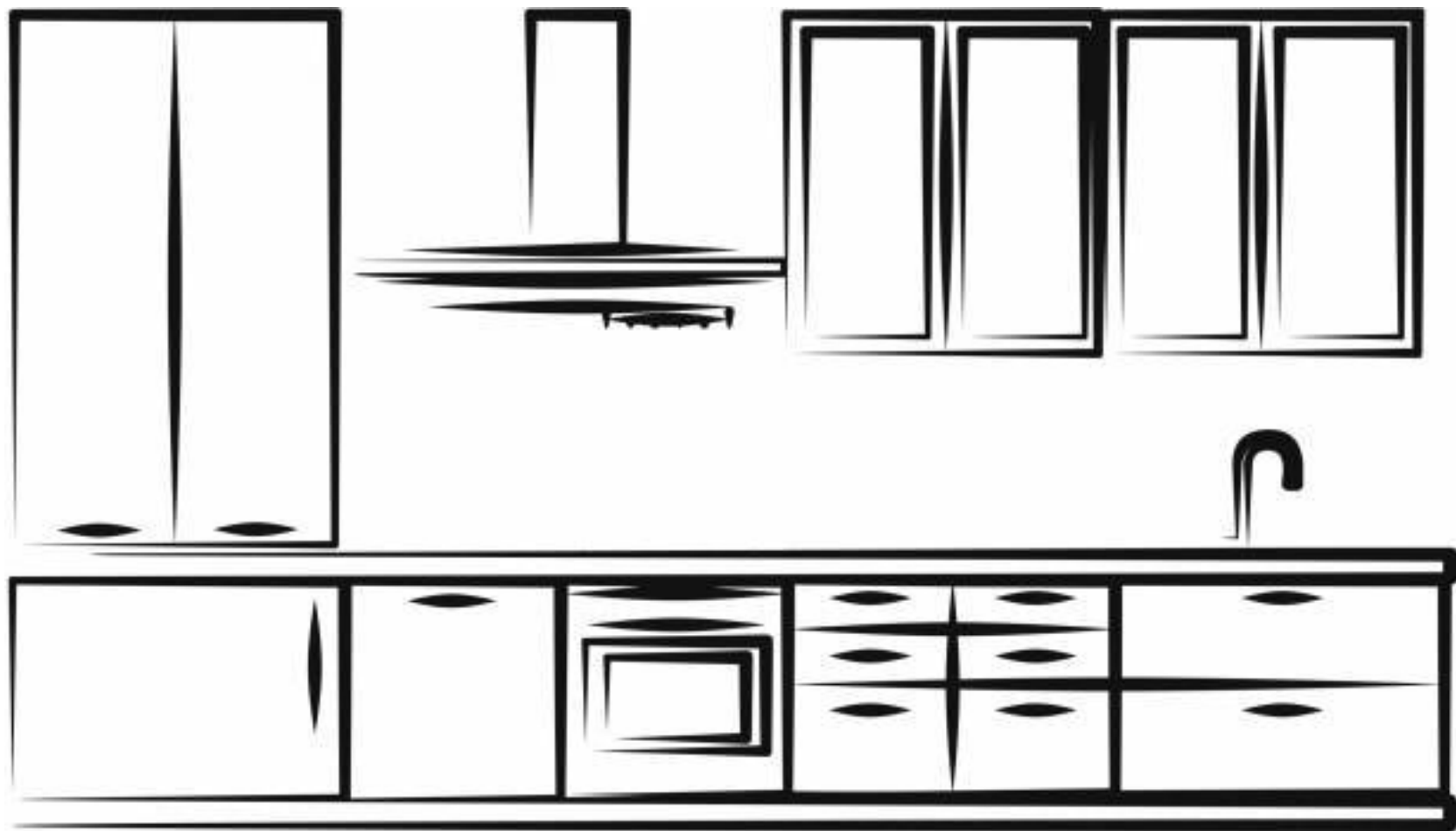
Command line:

`conda install beautifulsoup4` or `pip install beautifulsoup4`

BUT WAIT!



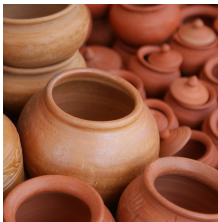
Python



Python + built-in modules



Python + built-in modules + self-made modules



Python + built-in modules + special packages

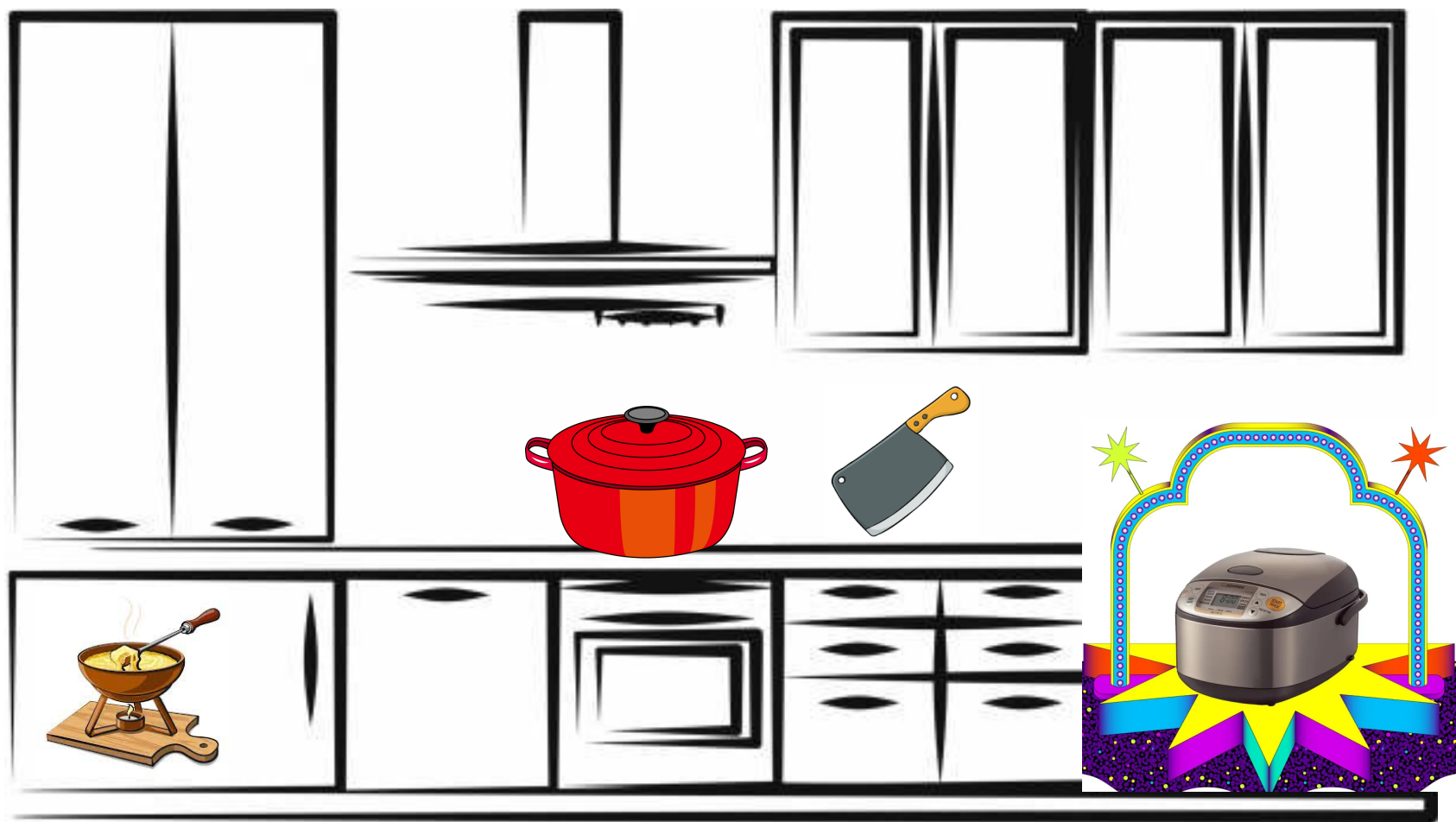


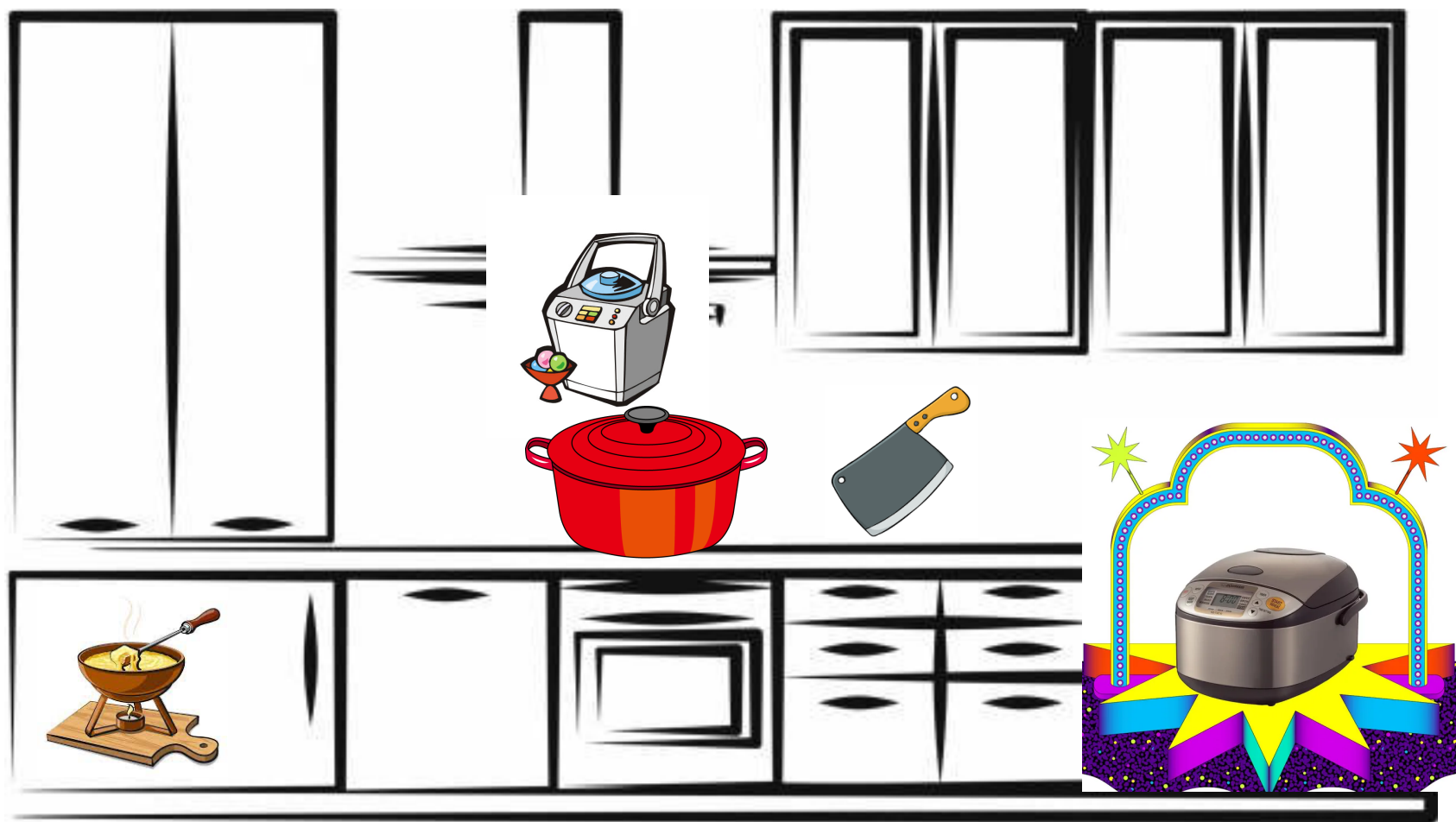
Python + built-in modules + special packages

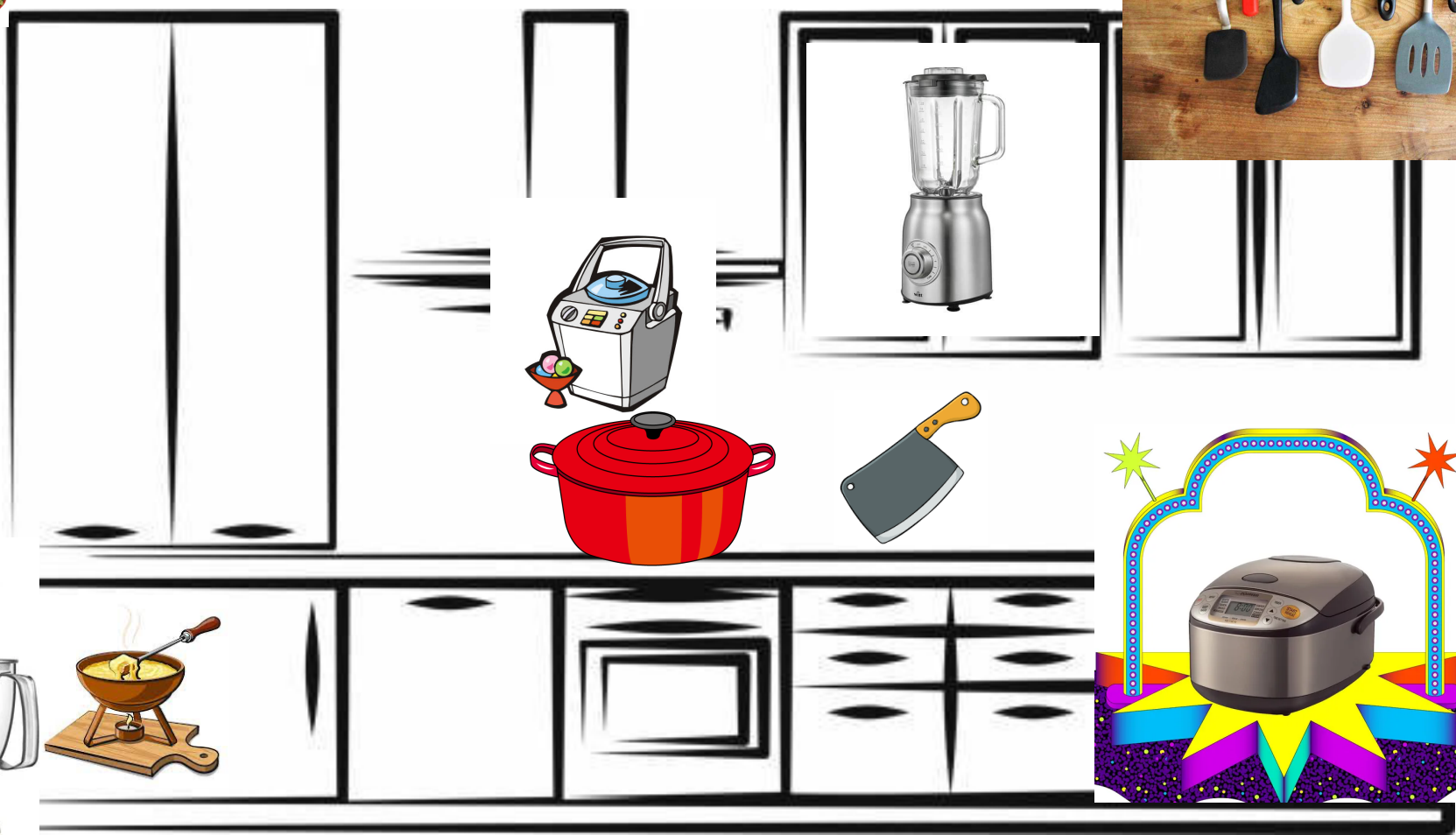


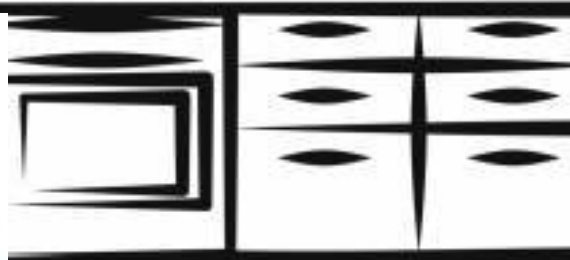
Installing new packages...



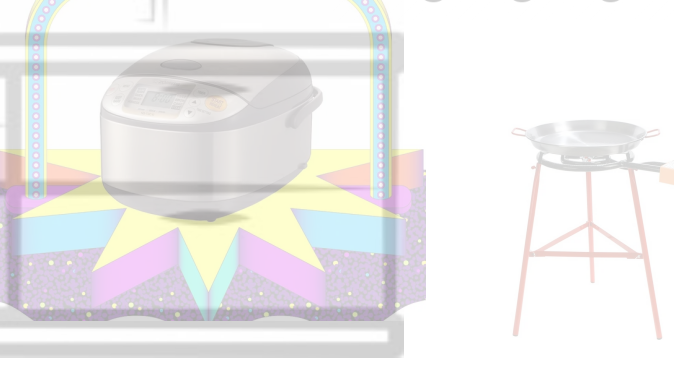
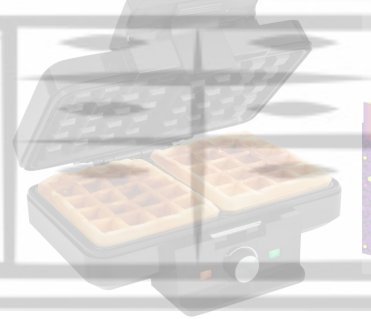
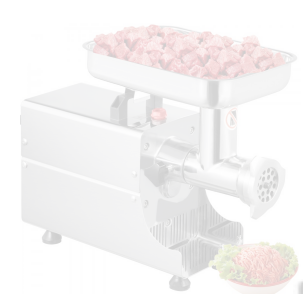




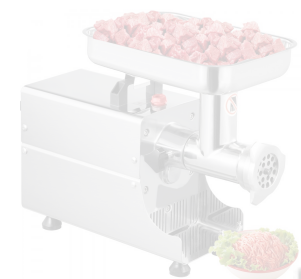




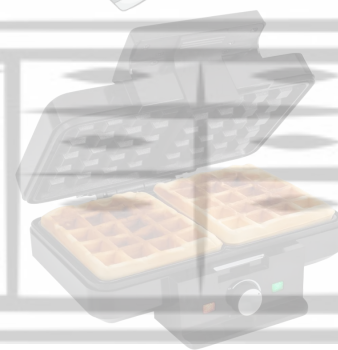
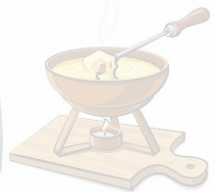


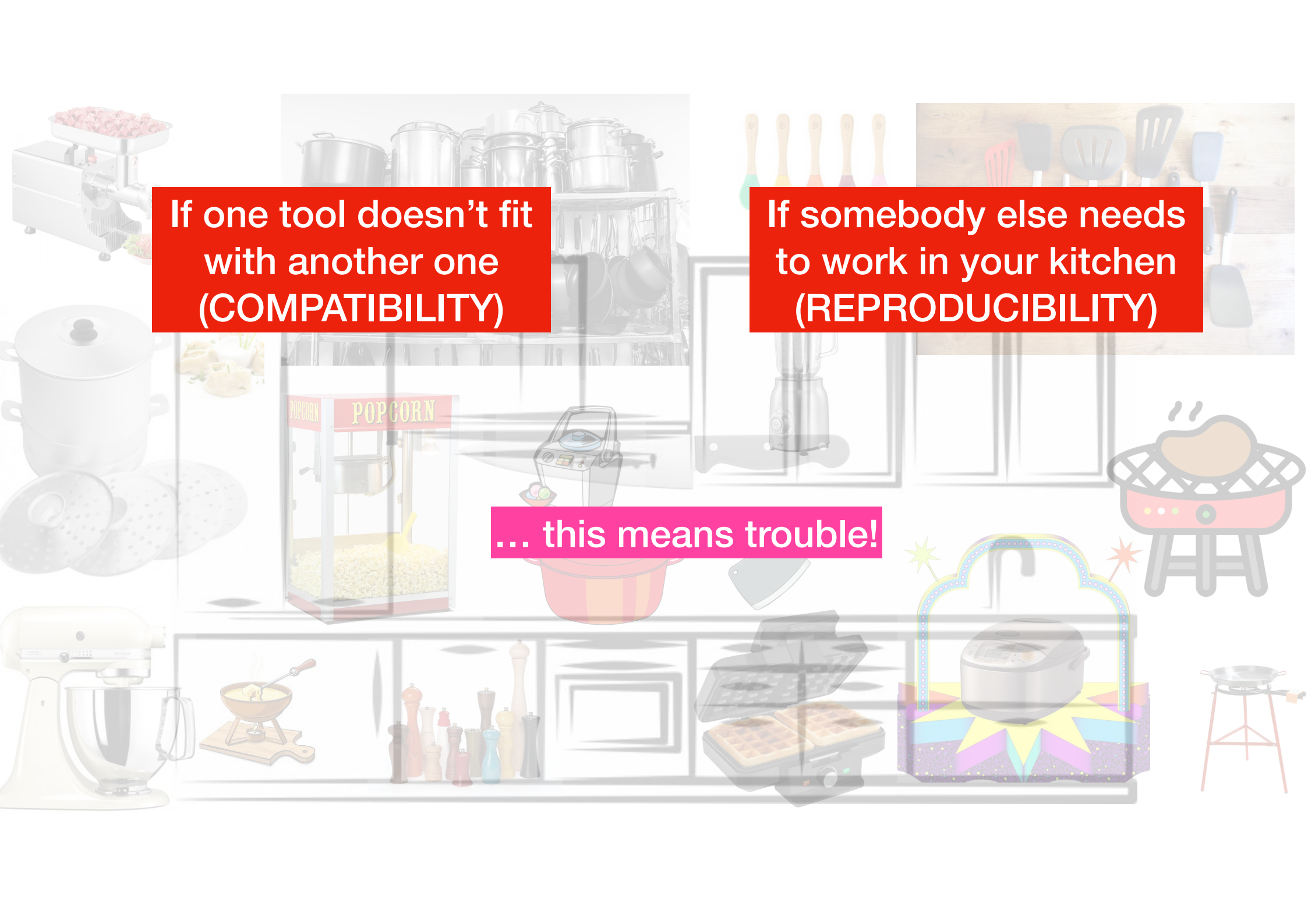


This is what you end up with
if you install everything in the same place....



... this means trouble!

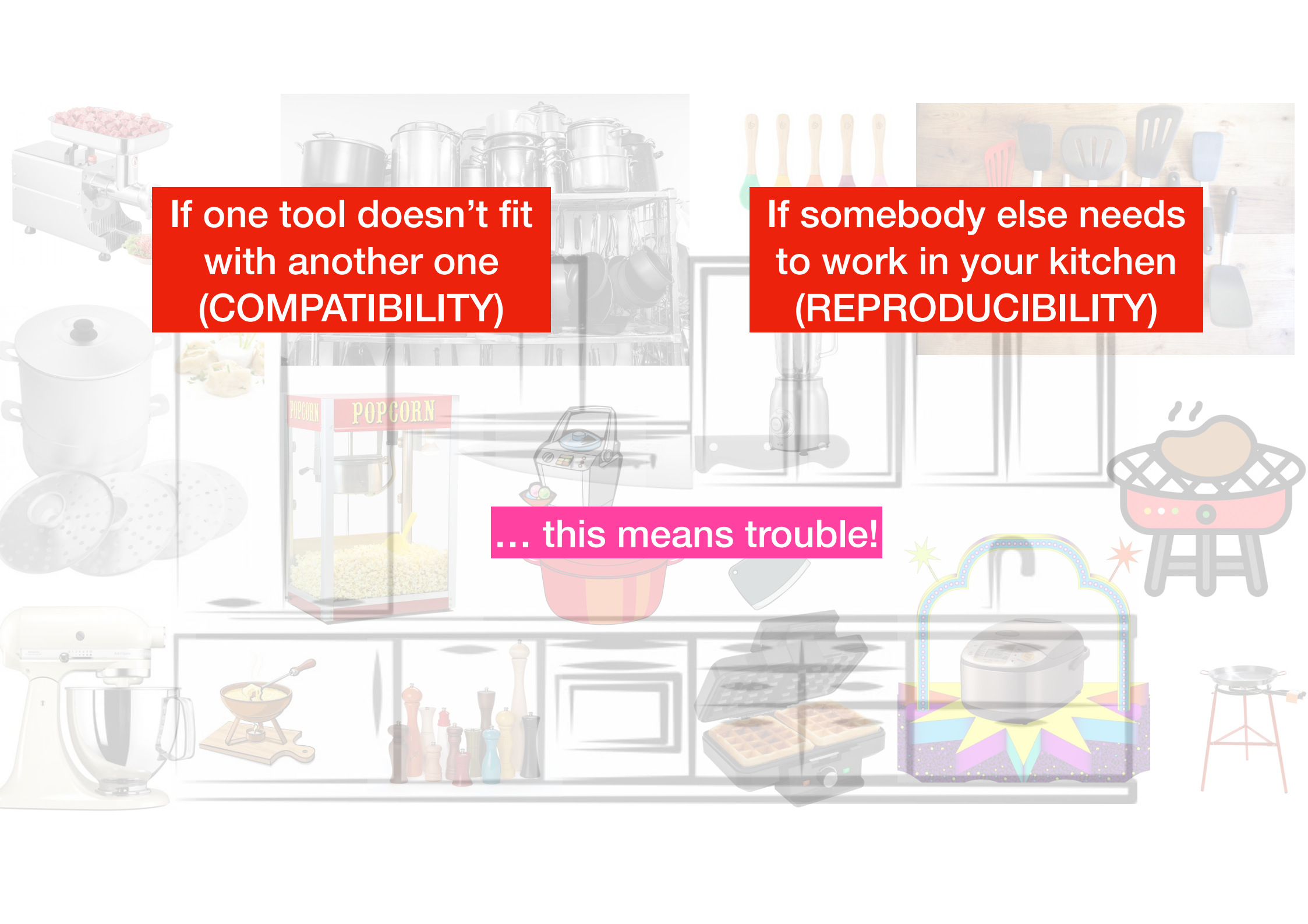


A collage of various kitchen appliances and tools, including a meat grinder, a large pot, a popcorn machine, a stand mixer, a waffle maker, a rice cooker, a grill, and various utensils. Two red text boxes are overlaid on the image. The first red box on the left contains the text "If one tool doesn't fit with another one (COMPATIBILITY)". The second red box on the right contains the text "If somebody else needs to work in your kitchen (REPRODUCIBILITY)". A central pink text box contains the text "... this means trouble!".

If one tool doesn't fit with another one (COMPATIBILITY)

If somebody else needs to work in your kitchen (REPRODUCIBILITY)

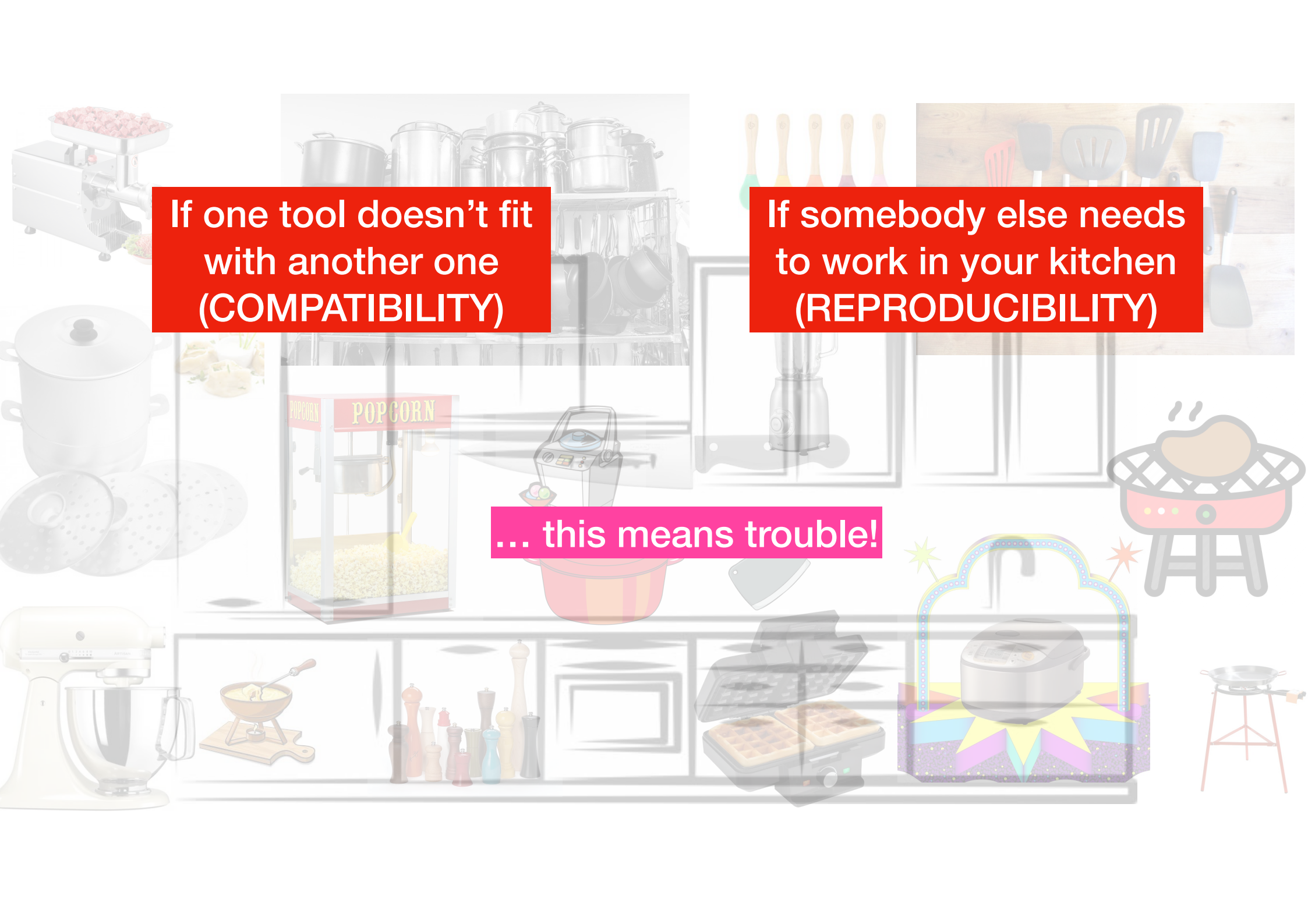
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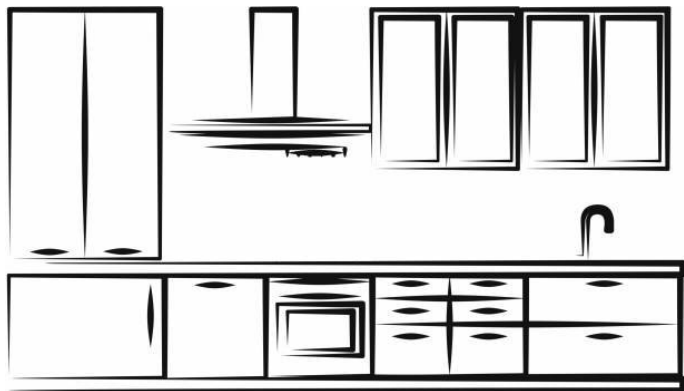
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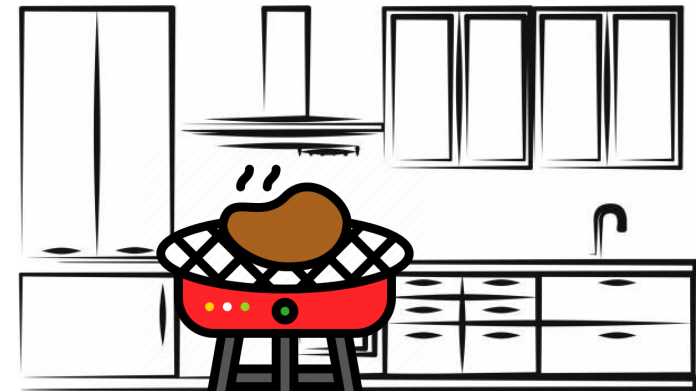
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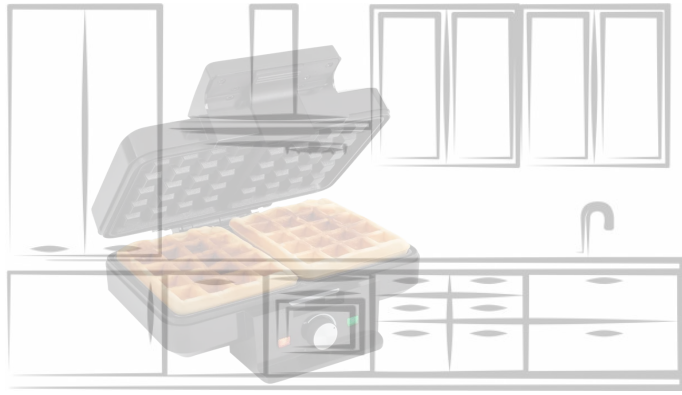
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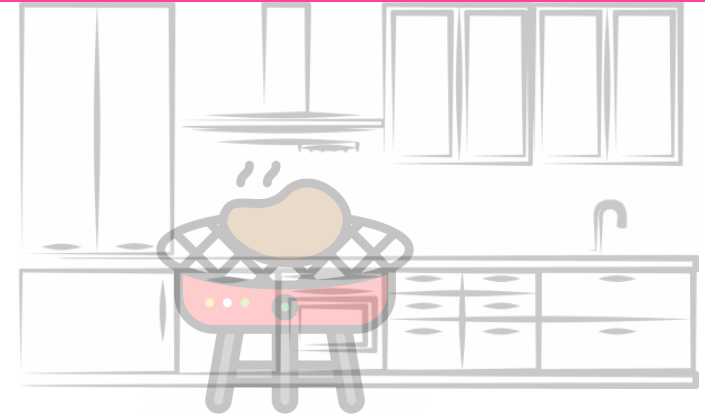
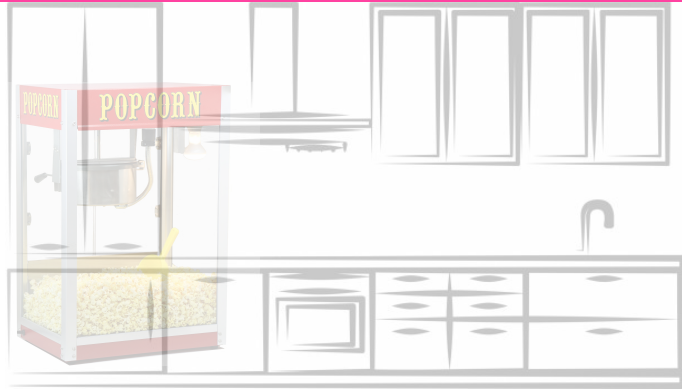
Every project has its own **ENVIRONMENT**



Every project has its own **ENVIRONMENT**



Python + only the packages you need for this specific project.

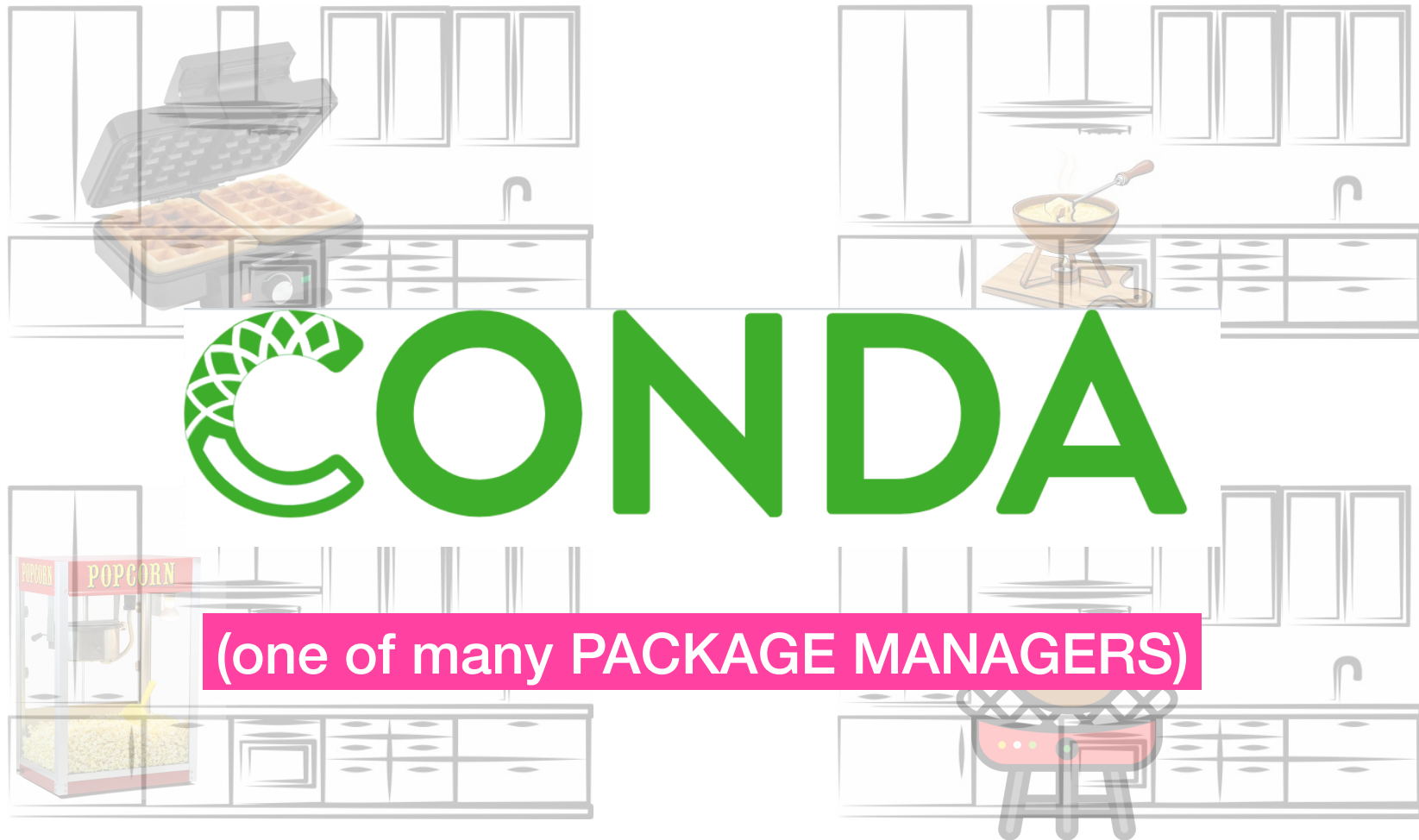


Every project has its own **ENVIRONMENT**



Compatibility &
Reproducibility:
Not solved, but
MUCH EASIER!

Every project has its own **ENVIRONMENT**



Environments with conda

From the command line

- `conda list` List all packages in CURRENT environment
- `conda env list` List all environments
- `conda create --name <environmentname>` Create a new environment
- `conda activate <environmentname>` Activate the new environment
- `conda install <packagename>` Install a new package in the CURRENT environment
- `conda deactivate` “Get out of” the new environment (back to base)
- `conda remove --name <environmentname> --all` Remove an environment